

Last-Mile Delivery Optimization Using Simulation and MILP

Lyxa Case Study

An integrated engineering framework combining simulation and optimization

Lebanese American University
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Focus

Smart last-mile logistics combining simulation and optimization.



Simulation and Optimization Driven Dispatching

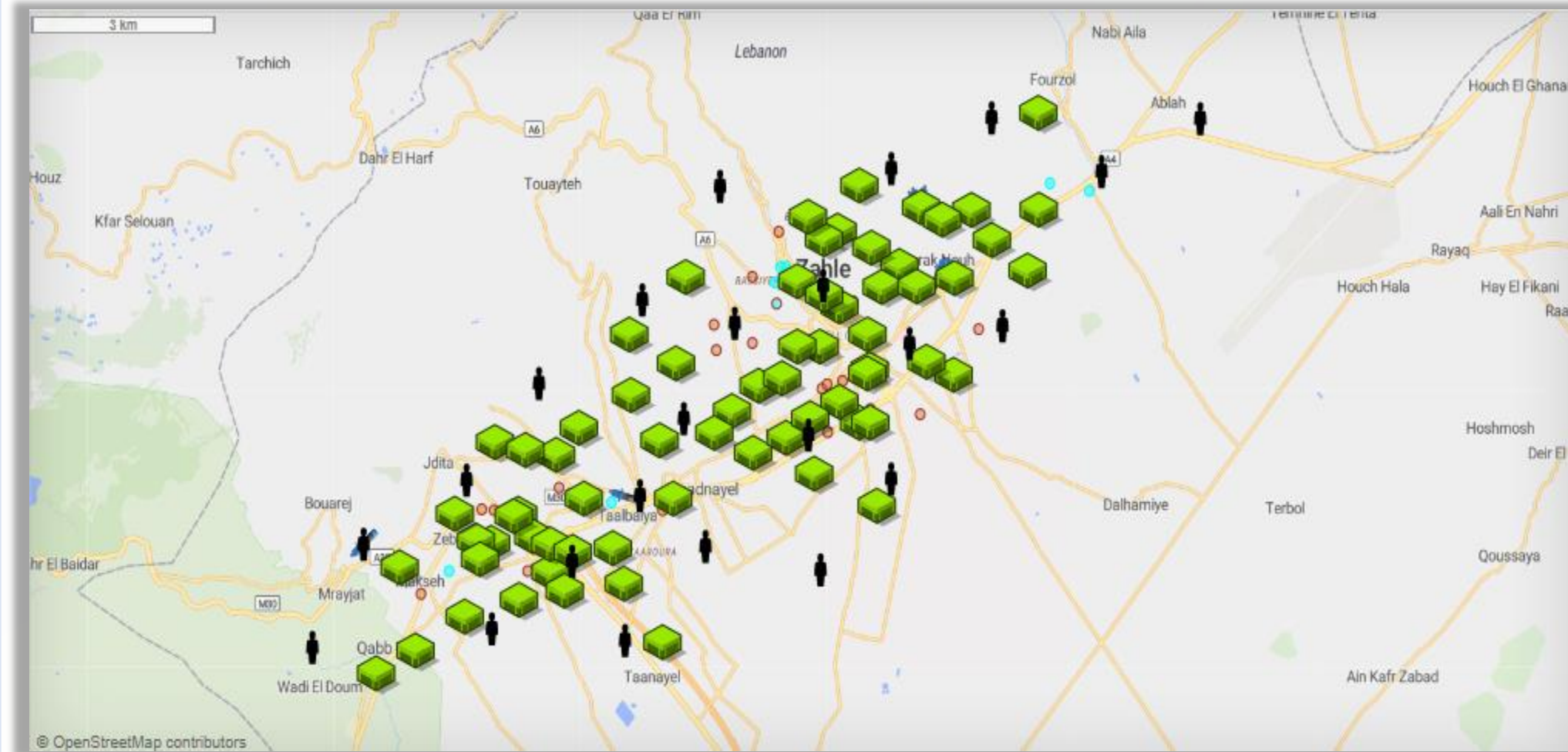
Key Features:

- Discrete event simulation using AnyLogic
- X riders, Y restaurants, Z customer clusters
- 45-minute Service Level Agreement (SLA) target
- Distance cost: \$v per km | Fixed rider cost: \$f

Two Strategies Compared:

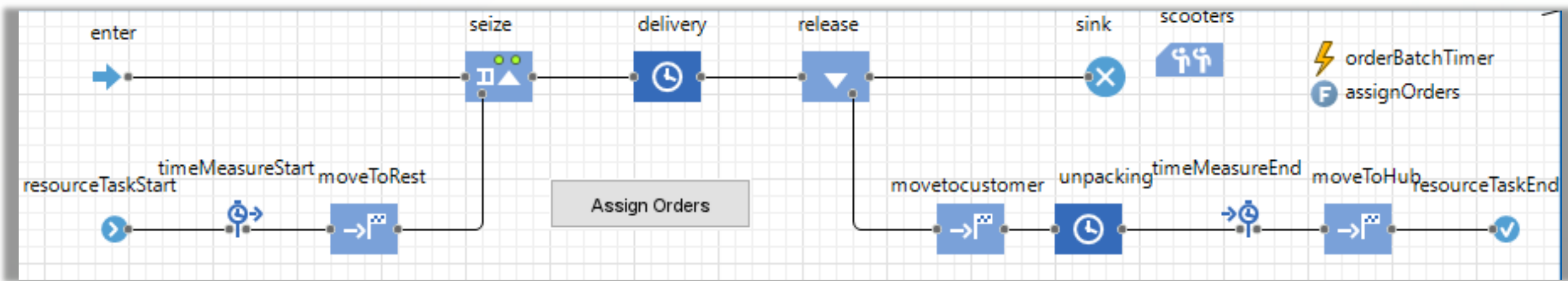
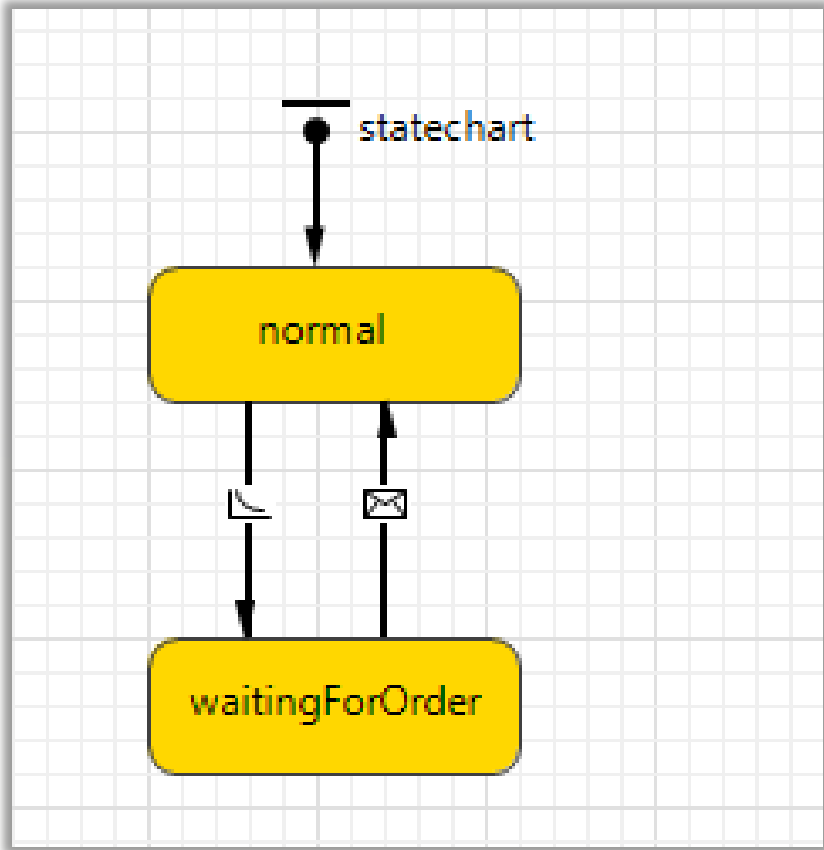
Baseline: Random assignment

Optimized: MILP assignment solver
(using OR Tools)

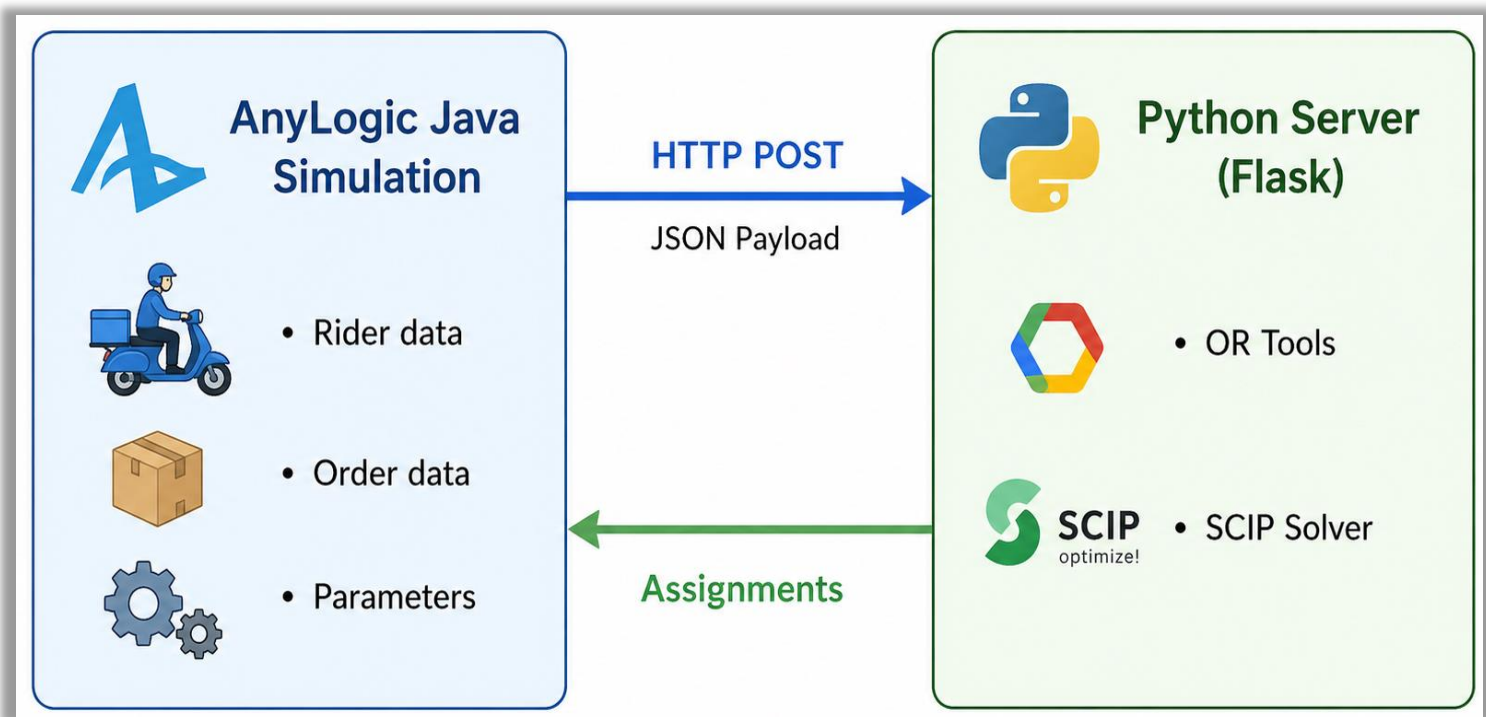


Simulation Agents

Agent	What It Does
Customer	Generates orders and waits for delivery
Restaurant	Prepares food for pickup
Rider	Delivers orders from restaurant to customer
Order	Travels through the process from creation to delivery



Optimization Engine



3. Objective Function

Minimize Total Cost = $\sum (\$1.20 \times Y_i) + \sum (\text{distance} \times \$0.22 \times X_{\{i,j\}})$

1. Decision Variables

Variable	Meaning	How Many
$x_{\{i,j\}}$	1 if rider i is assigned to order j , else 0	riders $\times$ orders
y_i	1 if rider i is used in this batch, else 0	one per rider

2. Constraints

Constraint	Meaning	Mathematical Form
Assignment	Every order must go to exactly one rider	$\sum_i x_{\{i,j\}} = 1$
Capacity	Each rider takes at most 1 order per batch	$\sum_j x_{\{i,j\}} \leq 1$
Linking	If a rider takes an order, they are used	$\sum_j x_{\{i,j\}} \leq M \times y_i$
Feasibility	Delivery must be possible before deadline	$\text{arrival_time} \leq \text{deadline}$

Results

Each simulation was run 10 times with different random seeds.

Metric	Baseline	Optimized	Improvement
Average Orders Completed	118	121	3 orders difference
Average Total Distance	698.42 km	589.67 km	↓ 108.75 km (15.6%)
Average Distance Per Order	5.92 km	4.87 km	↓ 1.05 km (17.7%)
Average Grand Total Cost	\$301.18	\$274.073	↓ \$27.1062(9%)
Average Cost Per Order	\$2.55	\$2.26	↓ \$0.29 (11.4%)
Average Delivery Time	27.8 minutes	25.9 minutes	↓ 1.9 minutes (6.8%)
Average SLA Compliance	80.5%	91.2%	↑ 10.7 percentage points

Factor	Explanation
Global View	The MILP solver considers all orders simultaneously instead of one by one
Shorter Routes	The solver identifies better assignments, reducing distance by an average of 108.75 km per run
Deadline Awareness	The solver respects delivery deadlines in its deterministic calculations, improving SLA compliance from 80.5% to 91.2%



Conclusion

This project demonstrated that a MILP-based dispatching optimizer consistently outperforms a greedy "nearest rider" heuristic in last-mile delivery operations.



What Was Delivered

Consolidating orders onto fewer riders increases distance slightly but reduces fixed costs, lowering total operational cost by 9% while cutting delivery times by 1.9 minutes.



Future Work

Multi-order batching, live traffic integration, and predictive prep time modeling.



Broader Impact

Economic savings, reduced emissions, improved customer experience, and applicability across on-demand logistics.

- ✓ Framework applies beyond food delivery to any on-demand logistics (e-commerce, courier, healthcare)